Isolation and Identification of *Bacillus* spp. and Related Genera from Different Starchy Foods

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ABSTRACT: Samples of butternut squash, potatoes, rice, and wheat flour were analyzed. *Bacillus* spp. and related species belonging to *Paenibacillus* and *Brevibacillus* genera were found in 96% of the samples. In butternut squash, predominant species were *Bacillus pumilus* and *Paenibacillus polymyxa* together with other *Bacillus* spp. species (*B. cereus, B. licheniformis, B. sphaericus,* and *B. subtilis*). In all the potato samples, *Bacillus* species were detected (*B. cereus, B. mycoides,* and *B. licheniformis*). Also, *Bacillus* spp. were detected in 100% of the unhusked rice samples, while incidence in white rice samples was 83%. In total rice samples, *B. pumilus, Brevibacillus brevis,* and *Paenibacillus macerans* were the main species and *B. cereus, P. polymyxa, B. subtilis,* and *Brevibacillus laterosporus* had the lower percentage. The most important species found in wheat flour was *P. polymyxa* with colony forming units per gram of about 10^2. As the identified species were potentially causatives of foodborne diseases, attention should be given to sanitary and temperature conditions as critical factors that influence the safety and shelf life of these products.

Practical Application: Foodborne illness produce by *B. cereus* have been associated with a wide variety of food. In addition, some other *Bacillus* species have been related to foodborne disease in humans. Information about the virulence mechanisms of other *Bacillus* spp. is scanty and their risk is underestimated. Identifying the group of food and the food processes in which *Bacillus cereus* or other *Bacillus* spp. would be hazardous for human health is vital for the prevention of foodborne outbreak. In this study, we determined the incidence of *Bacillus* spp. and related genera in some food items of agriculture origin from Argentina. This research is relevant to identify the presence of potentially pathogen *Bacillus* species and related genera in this type of food.

Keywords: *Bacillus subtilis*, bacterial spores, food microbiology, food safety

Introduction

Members of the aerobic spore-forming genus *Bacillus* spp. and other closely related species can be recovered from almost every environment in the biosphere. They are frequently isolated from both the natural habitat (soil and growing plants) and foods, including fresh vegetables (Ponce and others 2002; Sessitsch and others 2003) and minimally processed vegetables (Carlin and others 2000; Nguz and others 2005; Guinebretiere and others 2003), cereals and cereal by-products (Sarrías and others 2002; Finlay and others 2002; Sorokulova and others 2003; Haque and Russell 2005; From and others 2007), spices (Rusul and Yaacob 1995; Banerjee and Sarkar 2003), raw and pasteurized milk and milk products (Reyes and others 2007), raw meat and meat products (Abostate and others 2006; Uzeh and others 2006), and other food products such as honey, cheeses, anchovies, spices, and so on (Iurlina and others 2006).

Bacillus spp. and related genera have been associated with food spoilage such as ropy bread (Thompson and others 1998; Sorokulova and others 2003) besides causing several human infections that cause a range of diseases (Lequin and others 2005; Callegan and others 2006; Tena and others 2007), and incidents of foodborne illness (Kniehl and others 2003; Dierick and others 2005).

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Foodborne outbreaks caused by *Bacillus cereus* have been associated with almost all the categories of foodstuffs (EFSA 2005). Also some other *Bacillus* species, such as *Bacillus licheniformis* (Salkinoja-Salonen and others 1999), *Bacillus subtilis* (From and others 2005), *Bacillus pumilus* (Tena and others 2007), *Bacillus thuringiensis* (Jackson and others 1995), and so on, have been found responsible for diseases in human, including gastroenteritis, meningitis, endophthalmitis, and cutaneous infection.

The external surface of vegetables is contaminated with microorganisms from the soil, air, and animals. *Bacillus* spp. is a common soil and rhizosphere inhabitant (Reinoso Pozo and others 2006). Several strains have been isolated from different vegetables (Ponce and others 2002; Valero and others 2002; Rosenquist and others 2005).

Rice is usually stored with a moisture content range from 12% to 14%, and at this level, it is not possible for the vegetative forms of *Bacillus* spp. and other closely related species to proliferate (Haque and Russell 2005). Drying of grains destroys the majority of microorganisms, although spores are able to survive these conditions. Rice grains, even in the refined polished form, is a reliable source of *Bacillus* spp. with isolation rates ranging from 46% to 100% for *B. cereus*. When rice is cooked with water, milk, eggs, or other hydrating ingredients, a suitable environment for spore germination, growth of the organism, and toxin synthesis is produced (Haque and Russell 2005; From and others 2007).

Although flour is generally regarded as a microbiologically safe product due to its low water activity, all types of flour, especially wheat flour, are contaminated with *Bacillus* spp. as a result of soil contamination, cultivation, and processing methods (Sorokulova and others 2003). During milling, grinding and sifting

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operations create a considerable amount of heat, therefore moisture condensation in the break rolls, reduction rolls, and sifters can sometimes lead to buildup of flour residues inside equipment. Microorganisms may become established in these moisture residues, causing contamination of the mill products (Berghofera and others 2003).

Rice-based products and starchy foods such as pasta and potato are frequently contaminated and involved in *B. cereus* poisoning (Ankolekar and others 2009). Because of endospore formation, *Bacillus* spp. easily survive dried environments and heat treatments, being able to multiply in cooked food. Knowing the problems produced by species of *Bacillus* spp. and closely related genera, the presence of these species in different starchy foods has not been sufficiently investigated. The numbers of *B. cereus* in food that can cause illness are estimated to be higher than 10^4 to 10^5 g^{-1} of food. In Argentina, no information is available on the incidence of enterotoxigenic strains of *B. cereus* in foods and there is no sufficient information about the levels of *Bacillus* spp. in different starchy foods.

The aims of this study were to enumerate, isolate, and identify *Bacillus* spp. and closely related genera from different starchy foods.

Materials and Methods

Foods were analyzed for the presence of *Bacillus* spp. and other closely related genera. Test foods consisted of butternut squash (*Curcubita moschata*) (14 samples), potatoes (*Solanum tuberosum* subsp. *tuberosum*) (9 samples), raw rice (genus *Oryza L*) (12 samples), and wheat flour (*Triticum vulgare*) (20 samples). Butternut squash, potatoes, and unhusked rice were packed in aseptic polyethylene bags and were obtained from commercial establishments, whereas flour and white rice were obtained already packed from the industry. Samples were kept at 4 °C. All foods were analyzed within 24 h of purchase.

Ten grams of each sample were aseptically weighed and homogenized with 90 mL sterile Butterfield's phosphate-buffered dilution water (Butterfield 1932), using a Stomacher 400 Circulator Homogenizer for 2 min and heated for 10 min at 80 °C to destroy vegetative bacteria and fungi and to make easier the isolation of bacilli from spores that survive the heat treatment. Ten-fold serial dilutions of the liquid supernatant were prepared with the same diluent. One milliliter of the dilutions were mixed with molten (45 °C) mannitol–egg yolk–polymyxin (MYP) agar (FDA 1998) and poured into plates. After incubation at 37 °C for 24 to 48 h, the colonies that appeared on the selected plates were counted as colony forming units (CFU) per gram fresh weight sample. Morphologically, different colonies were selected from agar plates.

Bacterial identification was based on morphological and biochemical tests. Selected strains were examined for the presence of spores by microscopy after growth on nutrient agar at 37 °C for 2 to 5 d. Microscopic examination included Gram stain and Shaffer–Fulton stain to detect the presence and the position of spores, whether subterminal or central (Claus and Berkeley 1986). After microscopic examination, the purified colonies were stored at 4 °C. The biochemical tests used were: catalase test (FDA 1998); oxidase test (FDA 1998); nitrate reduction test (AOAC 1995); Voges Proskauer reaction (AOAC 1995); indole production test (Claus and Berkeley 1986); growth on Simmons citrate agar (AOAC 1995); growth with lysozyme (AOAC 1995); sugar fermentation test (Claus and Berkeley 1986); motility (AOAC 1995); hydrolysis of starch, gelatin (Claus and Berkeley 1986), and casein (Claus and Berkeley 1986); anaerobic growth (Claus and Berkeley 1986). In addition, the isolates were characterized by biochemical reactions in the API 50 CH kits at 30 °C according to the manufacturer's instructions (API System, BioMérieux S.A., Marcy l'Etoile, France).

Results and Discussion

The presence of *Bacillus* spp. and other closely related species in the different starchy food products is showed in Figure 1. A total of 55 samples from butternut squash, potato, rice, and wheat flour were analyzed. A number of 53 samples resulted positive to *Bacillus* spp. and related genera, which represented almost 96% of the total sample.

Soil is the primary source of food contamination with spores of *Bacillus* spp. and other closely related species that can contain between 10^2 and 10^5 spores per gram (Garbeva and others 2004).

In this study, 47.8% of raw vegetables samples (butternut squash and potato) were positive for *B. cereus*. Similar results were reported by Valero and others (2002), who found 51.8% positive samples. *B. cereus*-like bacteria was found in fresh fruits and vegetables such as cucumbers and in starchy ingredients (Valero and others 2002; Rosenquist and others 2005). Other species, such as *B. subtilis* and *B. pumilus* were found in Swiss chard by Ponce and others (2002).

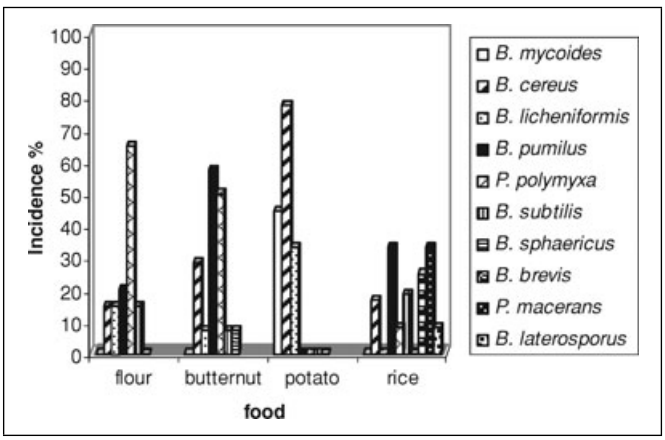


Figure 1 — Positive results of *Bacillus* spp. expressed as percentage (%) of samples containing *Bacillus* and related species isolated on butternut squash, potato, rice, and flour.

Table 1 — *Bacillus* spp. counts in different raw vegetable materials.

Food	Nr of samples	Microorganism	Range of counts on MYP (CFU/g)
Butternut squash	14	*B. cereus*	1×10^2
		B. licheniformis	1×10^2
		P. polymyxa	1×10^2 to 3×10^2
		B. pumilus	1×10^2 to 2×10^2
		B. sphaericus	1×10^2
		B. subtilis	1×10^2
Potato	9	*B. cereus*	1×10^2 to 7×10^2
		B. licheniformis	1×10^2 to 2×10^2
		B. mycoides	1×10^2 to 2.8×10^3
Rice	12	*B. brevis*	1×10^2 to 9×10^2
		B. cereus	1×10^2 to 3×10^2
		B. laterosporus	1×10^2
		P. macerans	1×10^2 to 7.7×10^3
		P. polymyxa	1×10^2
		B. pumilus	1×10^2 to 5×10^2
		B. subtilis	1×10^2 to 2.7×10^3
Flour	20	*B. cereus*	1×10^2 to 2×10^2
		B. licheniformis	1×10^2
		P. polymyxa	1×10^2 to 4×10^2
		B. pumilus	1×10^2 to 2×10^2
		B. subtilis	1×10^2 to 2×10^2

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Table 2—Percentage of samples containing *Bacillus spp.* from white and unhusked rice.

	Percentage of samples containing *Bacillus* spp.						
Rice	***B. cereus***	***B. polymyxa***	***B. laterosporus***	***B. pumilus***	***B. macerans***	***B. brevis***	***B. subtilis***
White	16.7	16.7		16.7	50	50	16.7
Unhusked	16.7		16.7	50	16.7		33.3

Table 1 shows *Bacillus* spp. counts in the different raw vegetables analyzed. Butternut squash presented a contamination level of about 10^2 CFU/g. By applying a restricted number of biochemical tests for the differential characters, Claus and Berkeley (1986) were able to identify *Bacillus* spp. and related species and also suggested the utilization of the API system, with some morphological and supplementary tests to achieve the classification. The identification tests and the carbohydrate patterns (API50CHB) of the isolated colonies confirmed that *B. pumilus* and *Paenibacillus polymyxa* were the predominant species (Figure 1) with an incidence of 57.1% and 50%, respectively. Besides these, other species with less positive cases were found such as *B. cereus*, *B. licheniformis*, *B. sphaericus*, and *B. subtilis*.

Bacillus spp. was present in all the potato samples analyzed (9). In total, 3 different species were found: *B. cereus*, *B. mycoides*, and *B. licheniformis*. *B. cereus* was the predominant species and was isolated from 77.7% of potato samples.

Both butternut squash and potato have a high content of dry matter starch and are often employed in the preparation of baby food. These foods are usually processed with heat treatment, insufficient to destroy the spores present in raw vegetables. The growth of bacteria during food storage will depend on storage temperatures and on the initial contamination of raw vegetables. *Bacilus circulans*, *Paenibacillus macerans*, and *P. polymyxa* have been reported to be the dominant species in vegetables purees stored at refrigeration temperatures (Carlin and others 2000). On the other hand, potential human pathogens such as *B. subtilis*, *B. licheniformis*, *B. amyloliquefaciens*, *B. pumilus*, and *B. cereus* have been considered the dominant species in the same products stored at abuse temperatures (Carlin and others 2000).

Table 1 and Figure 1 show the incidence and counts of species of *Bacillus* and closely related genera found in the total of rice samples. Table 2 shows positive samples (%) of *Bacillus* spp. in white and unhusked rice. *Bacillus*-like species were found in 100% of the unhusked rice sample. Alternatively, 83.3% of white rice samples show the presence of *Bacillus* species. In addition, the values of CFU per gram of unhusked rice samples (10^3 CFU/g) were higher than those observed in white rice (10^2 CFU/g). Sarrías and others (2002) also reported higher *Bacillus* spp. counts in unhusked rice than in white rice.

In this study, the main species found in rice samples were *B. pumilus*, *Brevibacillus brevis*, and *P. macerans*, which are present in 33.3% of samples (Figure 1). Cottyn and others (2001) reported that *B. pumilus* was the main species detected in rice (54.54% of samples), detecting also the presence of *Brevibacillus brevis* and *P. macerans*. Luo and others (2005) also detected this species in rice plants and grains. Incidence of *B. cereus* in our rice samples was about 16.7% (Figure 1). From and others (2007) also reported *B. cereus* counts (<100 CFU/g) and high *B. pumilus* counts ($\sim 10^5$ CFU/g) in rice samples. *P. polymyxa*, *B. subtilis*, and *Brevibacillus laterosporus* were also detected in the rice samples analyzed. Cottyn and others (2001) also reported the presence of these species in rice raw material. Table 2 shows that *P. polymyxa* and *B. brevis* were present only in white rice. This difference may be due to the unknown origin of the samples or due to the presence of different botanic varieties for white rice and unhusked rice. Moreover, the high percentage of positive samples found for *P. macerans* could be explained in the same manner. On the other hand, the absence of *B. laterosporus* in white rice could indicate that this species grows only by sticking to the cereal hull.

Generally, husking operation eliminates most external microflora of rice, whereas boiling further decreases the microbial counts in white or commercial rice (Sarrías and others 2002). *Bacillus* spp. and other closely related genera present in rice grains were able to survive in cooked condition and even produce diseases. About 95% of emetic food-poisoning outbreaks are due to the consumption of rice (Kramer and Gilbert 1989). Moreover, Ji and others (2007) reported the presence of *B. brevis* and *B. subtilis* in a kind of cake prepared from rice powder and sticky rice powder.

Bacillus spp. counts obtained from flour samples were about 10^2 CFU/g. The species found in flour samples were *B. cereus*, *B. licheniformis*, *P. polymyxa*, *B. pumilus*, and *B. subtilis*. These results are in agreement with those reported by Sorokulova and others (2003) and Iurlina and others (2006), although the percentage of positive samples of each species in the total population of *Bacillus* spp. was different. Although in our study the main species were *P. polymyxa* (65%) and *B. pumilus* (20%), for Sorokulova and others (2003) were *B. subtilis* and *B. licheniformis*, and for Iurlina and others (2006) were *B. subtilis* (55%) and *B. pumilus* (45%) with a lower presence of *B. licheniformis*, *P. polymyxa*, *B. lentus*, and *B. cereus*. Alternatively, the values observed for *B. subtilis* (20%) were similar to those reported by Te Giffel and others (1996)(22%). In our study, the presence of *B. cereus* (15%) in flour samples was lower than those reported by Te Giffel and others (1996) (55%), and was in agreement with those found by Iurlina and others (2006) (11%).

During wheat cultivation, surface-adhering contaminants are concentrated in outer layers of the grain and from the wheat pass into the flour during milling (Kirschner and von Holy 1989). *B. cereus* and other *Bacillus* species are present in wheat and flour at low levels. The same species found in wheat flours have been also recovered from ropy bread (Sorokulova and others 2003). Bread is a principal food for people at all ages, including children, the elderly, and patients with various illness and it is considered unsafe for such people when bread with a high count of *Bacillus* strains with unknown properties is consumed (Sorokulova and others 2003).

Predominant species were different for every type of food analyzed. Although *P. polymyxa* was the predominant species in flour samples, *P. polymyxa* and *B. pumilus* were mainly found from butternut squash samples. *B. cereus* was the main species reported in potato. In rice, none of the species, which show an incidence higher than 50% were found, but several species presented between 20% and 40% of incidence.

Conclusion

The results of this study indicate that the presence of *Bacillus* spp. and related genera in starchy vegetable food should not be ignored, because of the spore-forming species may reside in the surface of vegetables, in the environment or in equipment, affecting the end product quality. Attention should be given to sanitary and temperature conditions as important factors, which influence the safety and shelf life of these products.

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